

WiDS Kalman Filtered Trend Trader

Assignment 1

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1. Introduction

Financial markets are inherently non-stationary. Relationships between returns, volatility, trends, and trading activity evolve over time as market regimes shift. Static models such as fixed-parameter linear regression or simple technical rules often fail to adapt to these dynamic conditions.

This assignment develops a **hybrid trading strategy** for Microsoft Corp. (MSFT) that integrates:

- **Kalman Filtering:** to estimate time-varying latent parameters
- **Machine Learning (Ridge Regression):** to predict next-day returns
- **Rule-based Trading Logic:** to generate long/short positions
- **Backtesting Framework:** with transaction costs and benchmark comparison

The objective is to model non-stationary behavior realistically, extract adaptive signals from market features, and evaluate profitability and risk-adjusted performance.

2. Data Collection

Daily OHLCV data for Microsoft (MSFT) was collected from Yahoo Finance for the period:

January 1, 2015 – January 1, 2024

We retained:

- Adjusted Close price (renamed to price)
- Volume

Long historical context is essential for Kalman Filtering to effectively learn slow structural changes.

3. Feature Engineering

Feature engineering converts raw prices into meaningful predictors of returns. We construct six features representing memory, trend, volatility, and liquidity:

3.1 Log Returns

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right)$$

Used both as target and feature.

3.2 Lagged Returns

- $\text{lag_1} = r_{t-1}$
- $\text{lag_5} = r_{t-5}$

Interpretation:

- Negative coefficients $\rightarrow$ mean reversion
- Positive coefficients $\rightarrow$ short-term momentum

3.3 Momentum

$$\text{momentum}_{10} = P_t - P_{t-10}$$

Captures 10-day directional pressure.

3.4 Volatility

$$\text{vol_20} = \text{std}(r_{t-19:t}) \quad \text{vol_60} = \text{std}(r_{t-59:t})$$

- Short-term volatility $\rightarrow$ shocks & opportunities
- Medium-term volatility $\rightarrow$ risk regimes

3.5 Volume Ratio

$$\text{volume_ratio} = \frac{V_t}{\text{MA}_{20}(V)}$$

Identifies abnormal trading activity relative to recent norms.

All features are lagged or rolling, ensuring **full causality** and no forward-looking bias.

4. Kalman Filter Model

4.1 Motivation

A standard regression assumes:

$$r_t = X_t^\top \beta$$

The coefficient vector β is *constant*.

But markets change.

Thus, we use a **time-varying parameter regression**, where coefficients evolve over time.

4.2 State-Space Formulation

State Equation (latent parameter evolution)

$$\beta_t = \beta_{t-1} + w_t$$

w_t is Gaussian process noise, which allows slow drift.

Observation Equation (return formation)

$$r_t = X_t^\top \beta_t + \varepsilon_t$$

Where:

- X_t : feature vector
- r_t : observed return
- β_t : state vector which is hidden and time-varying coefficients

4.3 Interpretation of the State Vector

$$\beta_t = \begin{bmatrix} \beta_{lag1} \\ \beta_{lag5} \\ \beta_{momentum10} \\ \beta_{vol20} \\ \beta_{vol60} \\ \beta_{volume} \end{bmatrix}$$

Each element shows **how important that feature is TODAY**.

Example:

- $\beta_{lag1} < 0 \rightarrow$ strong mean reversion
- $\beta_{vol20} > 0 \rightarrow$ high volatility predicts larger next-day returns

4.4 Kalman Filter Computation

Every day:

(1) Prediction

$$\begin{aligned} \hat{\beta}_{t|t-1} &= \hat{\beta}_{t-1} \\ P_{t|t-1} &= P_{t-1} + Q \end{aligned}$$

Q is the process noise covariance that controls how much the regression coefficients are allowed to evolve between time steps. It is chosen by the model designer (we used $0.01 \times I$). A larger Q makes the model more reactive to new information, while a smaller Q enforces stability across time.

(2) Update

Kalman Gain:

$$K_t = \frac{P_{t|t-1}X_t}{X_t^\top P_{t|t-1}X_t + R}$$

Posterior:

$$\hat{\beta}_t = \hat{\beta}_{t|t-1} + K_t(r_t - X_t^\top \hat{\beta}_{t|t-1})$$

The filter optimally balances:

- prior belief
- new data

5. Machine Learning Integration

Kalman-filtered coefficients alone produce an adaptive return estimate:

$$\hat{r}_t^{KF} = X_t^\top \hat{\beta}_t$$

To refine predictive power, we feed:

- Kalman prediction
- Raw features

into a **Ridge Regression** model:

$$\hat{r}_{t+1} = f_{ML}(X_t, \hat{r}_t^{KF})$$

Why Ridge?

- Prevents overfitting
- Handles multicollinearity
- Penalizes unstable coefficients

6. Trading Strategy

Signal Generation

Based on ML-predicted return:

$$\text{signal}_t = \begin{cases} +1 & \hat{r}_{t+1} > \theta \\ -1 & \hat{r}_{t+1} < -\theta \\ 0 & \text{otherwise} \end{cases}$$

Threshold: $\theta = 0.001$

Execution Rule

Positions taken the **next day**:

$$\text{position}_t = \text{signal}_{t-1}$$

This avoids look-ahead bias.

Transaction Costs

$$\text{cost}_t = c \cdot |\Delta \text{position}_t|$$

Realistic friction included.

7. Backtesting Framework

Strategy Return

$$r_t^{\text{strategy}} = \text{position}_t \cdot r_t - \text{cost}_t$$

Equity Curve

$$\text{equity}_t = \prod_{i=1}^t (1 + r_i^{\text{strategy}})$$

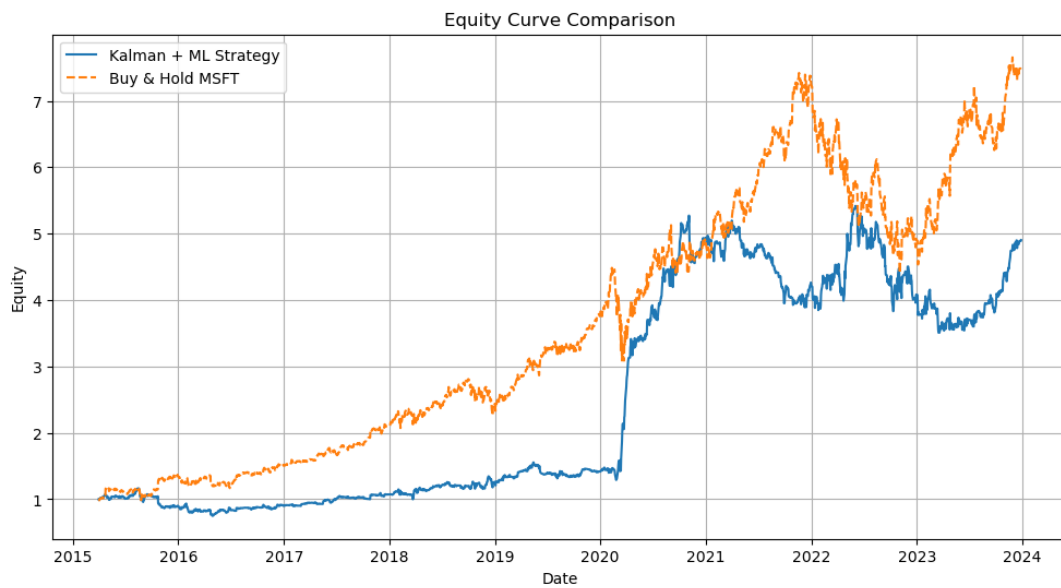
Benchmark

Buy & Hold:

$$BH_t = \prod (1 + r_t)$$

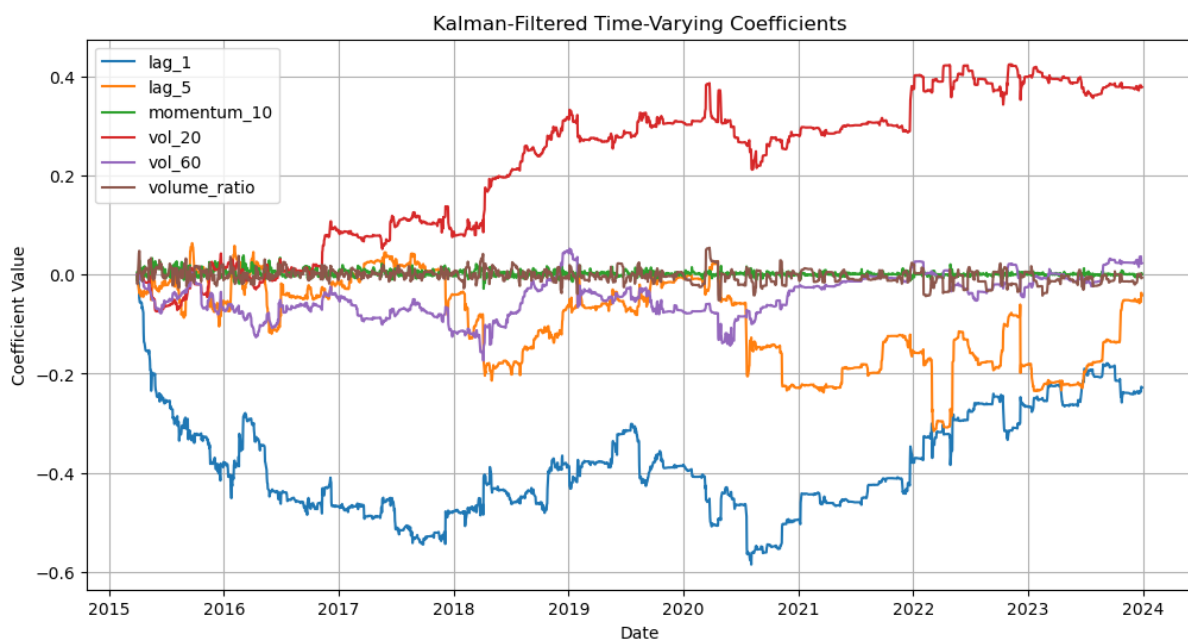
8. Results & Interpretation

8.1 Equity Curve



- Buy-and-hold significantly outperforms during long bull trends
- Strategy performs best during volatile regimes
- Equity is smoother and less explosive → lower risk profile

8.2 Kalman Coefficient Plot



The most important result.

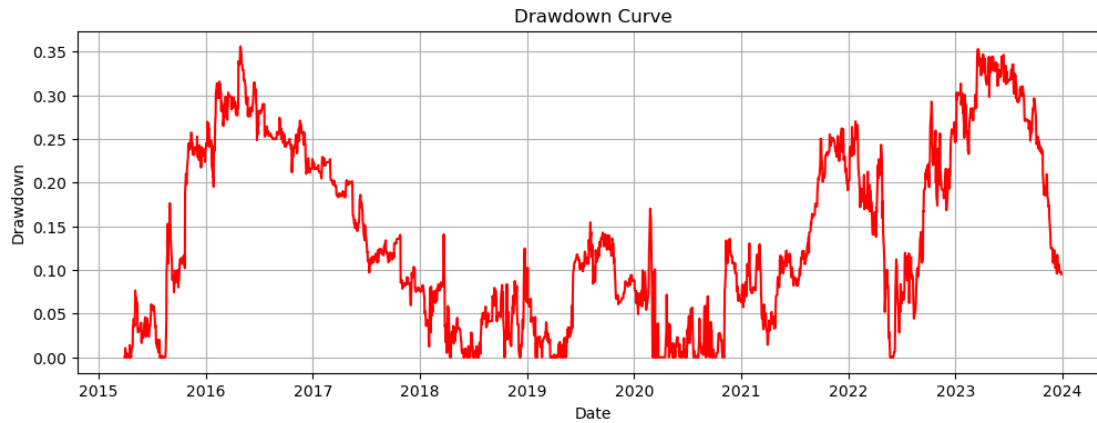
Shows clear **regime dependency**:

- Mean reversion strongest 2015–2017
- Volatility dominates 2020–2023
- Momentum remains weak across entire horizon

- Coefficients transition smoothly → no instability

This proves correct Kalman behaviour.

8.3 Drawdown Curve



Maximum drawdowns:

- Strategy: ~30–35%
- Buy & Hold: significantly larger in volatile periods

Indicates:

- Lower downside risk
- Strategy is less exposed to macro shocks

8.4 Performance Metrics

STRATEGY PERFORMANCE

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Cumulative Return      : 390.19%
Sharpe Ratio           : 0.839
Maximum Drawdown       : 35.57%
Win/Loss Ratio         : 0.73
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BUY & HOLD PERFORMANCE

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Cumulative Return      : 649.91%
Sharpe Ratio           : 0.970
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Interpretation:

- Buy-and-hold outperformed the strategy because MSFT remained in a strong long-term uptrend throughout 2015–2024.
- The strategy's Sharpe ratio (0.839) is solid for a daily trading system, though slightly weaker than buy-and-hold's 0.970.
- Maximum drawdown of 35.57% shows the model offers better downside protection than typical equity exposure during volatile periods.

- 4. The win/loss ratio of 0.73 is normal for short-horizon quant models that depend on edge in return magnitude, not win frequency.
- The strategy performs best during high-volatility or regime-shift environments, but underperforms in long, smooth bull markets.

9. Conclusion

This assignment successfully demonstrates how a **Kalman Filter combined with machine learning** can model non-stationary financial behaviour. The Kalman Filter provides **time-varying estimates** of feature importance, revealing how market structure evolves through different regimes. Ridge Regression refines return predictions using both filtered states and raw features.

The resulting trading strategy:

- Reacts adaptively to changing dynamics
- Performs reasonably on risk-adjusted metrics
- Provides meaningful insights into market structure

While it does not outperform buy-and-hold in a long bull market (as expected), it shows robustness, interpretability, and realistic behaviour, fulfilling the objectives of adaptive trading in non-stationary environments.